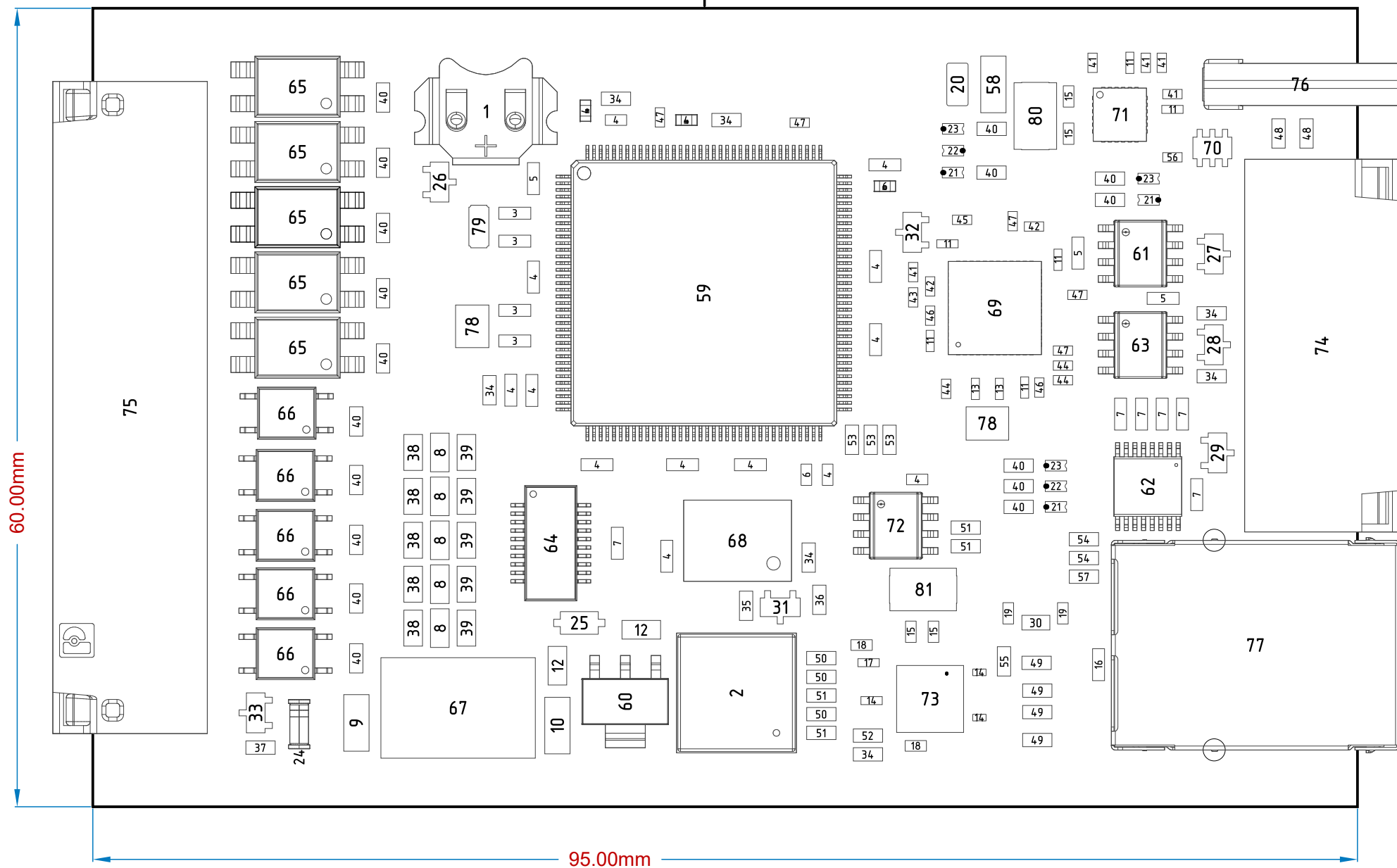


Вид спереди (2.5:1)



Bill Of Materials(1 cmp. u3 4)

Line #	Description	Comment	Designator	Footprint	Part	Alternative Part	Quantity
1	Battery	2998TR	BAT1	BAT_2998TR	2998TR		1
2	Piezoelectric Sounders	L-KLS3-SMT-09X1.8'	BUZ1	SPKR_PKMCS0909E4000-R1	L-KLS3-SMT-09X1.8	PKMCS0909E4000-R1	1
3	Capacitor	8pF 50V	C1, C2, C5, C6	C_0603	CC0603DRNPO9BN8R0		4
4	Capacitor	0.1uF 50V	C3, C4, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C40, C54, C56	C_0603	CC0603KPX7R9BB104		15
5	Capacitor	0.1uF 50V	C7, C21, C27	C_0603	CC0603KPX7R9BB104		3
6	Capacitor	2.2uF 16V	C8, C9	C_0603	CC0603KRX5R7BB225		2
7	Capacitor	0.1uF 50V	C22, C23, C24, C25, C26, C30	C_0603	CC0603KPX7R9BB104		6
8	Capacitor	2.2nF 50V	C28, C29, C31, C32, C33	C_0805	CC0805KRX7R9BB222	CL21B222KBAN NNC	5
9	Capacitor	10uF 50V	C34	C_1206	CC1206KKX5R9BB106	CL31A106KBHN NNE	1
10	Capacitor	22uF 25V	C35	C_1206	CC1206KKX5R8BB226	CL31A226KAHN NNE	1
11	Capacitor	100nF 16V	C36, C41, C42, C45, C46, C55	C_0402	CC0402KRX7R7BB104	CC0402KRX5R7BB104	6
12	Capacitor	10uF 10V	C37, C38	C_0805	CC0805KKX5R6BB106		2
13	Capacitor	8pF 50V	C43, C44	C_0402	CC0603DRNPO9BN8R0		2
14	Capacitor	0.01uF 50V	C47, C59, C60	C_0402	CC0402KRX7R9BB103		3
15	Capacitor	30pF 50V	C48, C53, C57, C58	C_0603	CC0603JRNPO9BN300		4
16	Capacitor	0.022uF	C49	C_0603	CC0603KRX7R9BB223		1
17	Capacitor	470pF 50V	C50	C_0402	CC0402KRX7R9BB471		1
18	Capacitor	1uF 50V	C51, C52	C_0603	CC0603KRX7R9BB105		2
19	Capacitor	1uF 50V	C61, C62	C_0603	CC0603KRX7R9BB105		2
20	Capacitor	1nF 1000V	C63	C_1206	CC1206KKX7RCBB102	CL31B102KIFNN NE	1
21	LED	Green	D1, D7, D11	LED_0603	FYLS-0603PGC	KP-1608CGCK	3
22	LED	Yellow	D2, D12	LED_0603	FYLS-0603PGC	KP-1608CGCK	2
23	LED	Red	D3, D5, D13	LED_0603	FYLS-0603PGC	KP-1608CGCK	3
24	Diode	BZV55C10	D4	SOD80C	BZV55C10		1
25	Diode	SM5819PL-TP	D6	SOD-123	SM5819PL-TP		1
26	Diode	BAT54	D8	SOT23	BAT54		1
27	ESD	ESDCAN24-2BLY	D9	SOT23	ESDCAN24-2BLY		1
28	ESD	SM712-02HTG	D10	SOT23	SM712-02HTG		1
29	Diode	ESD25L	D14	SOT23	ESDA25L		1
30	Inductor	1uH	L2	L_0603	LQM18FN1R0M00D		1
31	Transistor	BC847B	Q1	SOT23	BC847B		1
32	NPN Bipolar Transistor	MMBT9013	Q2	SOT23	MMBT9013		1

Bill Of Materials(2 cmp. u3 4)

Line #	Description	Comment	Designator	Footprint	Part	Alternative Part	Quantity
33	P-Channel MOSFET	SI2309A	Q6	SOT23	SI2309A		1
34	Resistor	10k	R1, R2, R5, R6, R31, R53, R78	R_0603	RC0603FR-0710KL		7
35	Resistor	1k	R3	R_0603	RC0603FR-071KL		1
36	Resistor	3.3k	R4	R_0603	RC0603FR-073K3L		1
37	Resistor	100k	R7	R_0603	RC0603JR-07100KL		1
38	Resistor	6.2k	R8, R10, R12, R14, R16	R_0805	RC0805JR-076K2L	CR-05JL7-6K2	5
39	Resistor	2.4k	R9, R11, R13, R15, R17	R_0805	RC0805JR-072K4L	CR-05JL7-2K4	5
40	Resistor	300	R18, R19, R20, R21, R22, R23, R24, R25, R26, R27, R32, R33, R61, R62, R63, R69, R70	R_0603	RC0603FR-07300RL		17
41	Resistor	10K	R28, R64, R65, R66, R67	R_0402	RC0402JR-0710KL		5
42	Resistor	100	R29, R37	R_0402	RC0402JR-07100RL		2
43	Resistor	36K	R30	R_0402	RC0402JR-0736KL		1
44	Resistor	4.7K	R34, R35, R38	R_0402	RC0402JR-074K7L		3
45	Resistor	1.5K	R36	R_0402	RC0402JR-071K5L		1
46	Resistor	100K	R39, R40	R_0402	RC0402JR-07100KL		2
47	Resistor	22	R41, R42, R43, R73, R74	R_0402	RC0402JR-0722RL		5
48	Resistor	5.1k	R44, R45	R_0603	RC0603JR-075K1L		2
49	Resistor	49.9	R46, R47, R48, R49	R_0603	RC0603FR-0749R9L		4
50	Resistor	10	R50, R57, R58	R_0603	RC0603FR-0710RL		3
51	Resistor	4.7k	R51, R71, R76, R77	R_0603	RC0603FR-074K7L		4
52	Resistor	33	R52	R_0603	RC0603FR-0733RL		1
53	Resistor	27	R54, R55, R56	R_0603	RC0402JR-0727RL		3
54	Resistor	330	R59, R72	R_0603	RC0603FR-07330RL		2
55	Resistor	12.1k	R60	R_0603	RC0603FR-0712K1L		1
56	Resistor	2.7K	R68	R_0402	RC0402FR-072K7L		1
57	Resistor	10k	R75	R_0603	RC0603FR-0710KL		1
58	Resistor	1M	R79	R_1206	RC1206FR-071ML		1
59	Microcontroller	STM32F765ZGT7	U1	TQFP144_M	STM32F765ZGT		1
60	LDO	LM1117GS-3.3	U2	SOT 223	LM1117GS-3.3		1
61	CAN_transiever	SN65HVD232D	U3	SO-8_M	SN65HVD232DR		1
62	ST232CTR	ST232CTR	U4	TSSOP16_STM	ST232CTR		1
63	RS485 Half Duplex Convertor	SN75176BDR	U5	SO-8	SN75176BDR		1
64	Isolated Octal Industrial Digital Input	CA-IS3988P	U6	SOP64P600X175-20N	CA-IS3988P		1

Bill Of Materials(3 cmp. u3 4)

Line #	Description	Comment	Designator	Footprint	Part	Alternative Part	Quantity
65	Solid State Relay 60V 3A	GAQY252G3EH	U7, U8, U10, U12, U15	SMD-4P	GAQY252G3EH		5
66	Solid State Relay 60V 250mA	GAQY221R2S	U9, U11, U13, U14, U16	SOP254P700X270-4N	GAQY221R2S		5
67	Non-isolated dc-dc	K7805-500R3	U17	K7805500R3	K7805-500R3		1
68	Flash Memory	W25Q128JVEIQ	U19	W-PDFN-8 (MLP8) – 8mm ? 6mm Se	W25Q128JVEIQ		1
69	STM32 ARM-based 32-bit MCU Performance Line with 128 kB Flash, 20 kB Internal RAM, -40 to +85°C Temperature, 48- Pin VFQFPN, Tray	STM32F103CBU6	U20	STM-UFQFPN48_M	STM32F103CBU6		1
70	17V Clamp 5A (8/20Âµs) Ipp Tvs Diode Surface Mount SOT-23-6	USBLC6-2SC6	U21	SOT95P280X145-6 N	USBLC6-2SC6		1
71	USB 2.0 HIGH SPEED 4-PORT HUB CONTROLLER	FE1.1S-BQFN24B	U23	QFN50P400X400X8 0-25N-D	FE1.1S-BQFN24B		1
72	256-Kbit I2C- compatible Serial EEPROM	24c256	U24	SO-8_M	AT24C256C		1
73	Phy Eth 2/2 Transceiver Full RMII 24-QFN-EP _4x4_	LAN8720A-CP-TR	U26	QFN50P400X400X1 00-25N	LAN8720A-CP-TR		1
74	THR-3.5-12P	1787056	X1	DMC 1.5/ 6- G1F-3.5-LR P20THR 1787056	1787056		1
75	THR-3.5-24P	1787111	X2	DMC 1,5/12- G1F-3,5- LRP20THRR72	1787111		1
76	USB Connector USB TYPE C UPRIGHT STRAIGHT TAB 14 PIN	KUSBX-SL-CS1N14-B	X3	MOLEX_216989000 1	KUSBX-SL-CS1N14-B		1

Bill Of Materials(4 cmp. u3 4)

Line #	Description	Comment	Designator	Footprint	Part	Alternative Part	Quantity
77	DIP RJ45 Connector;	HR915340A'	X4	RJ45-TH_HR915340A	HR915340A		1
78	Crystal Oscillator	NX3225GD	Y1, Y3	NX3225GD	NX3225GD-8MHZ-STD-CRA-3		2
79	Crystal Oscillator	NX3215SA-32.768K-STD-MUA-14	Y2	XTAL_NX3215SA-32.768K-STD-MUA-8	NX3215SA-32.768K-STD-MUA-14		1
80	Crystal Oscillator	X503212MSB2GI	Y4	X5032	X503212MSB2GI		1
81	Crystal Oscillator	X503225MSB2GI	Y5	X5032	X503225MSB2GI		1

